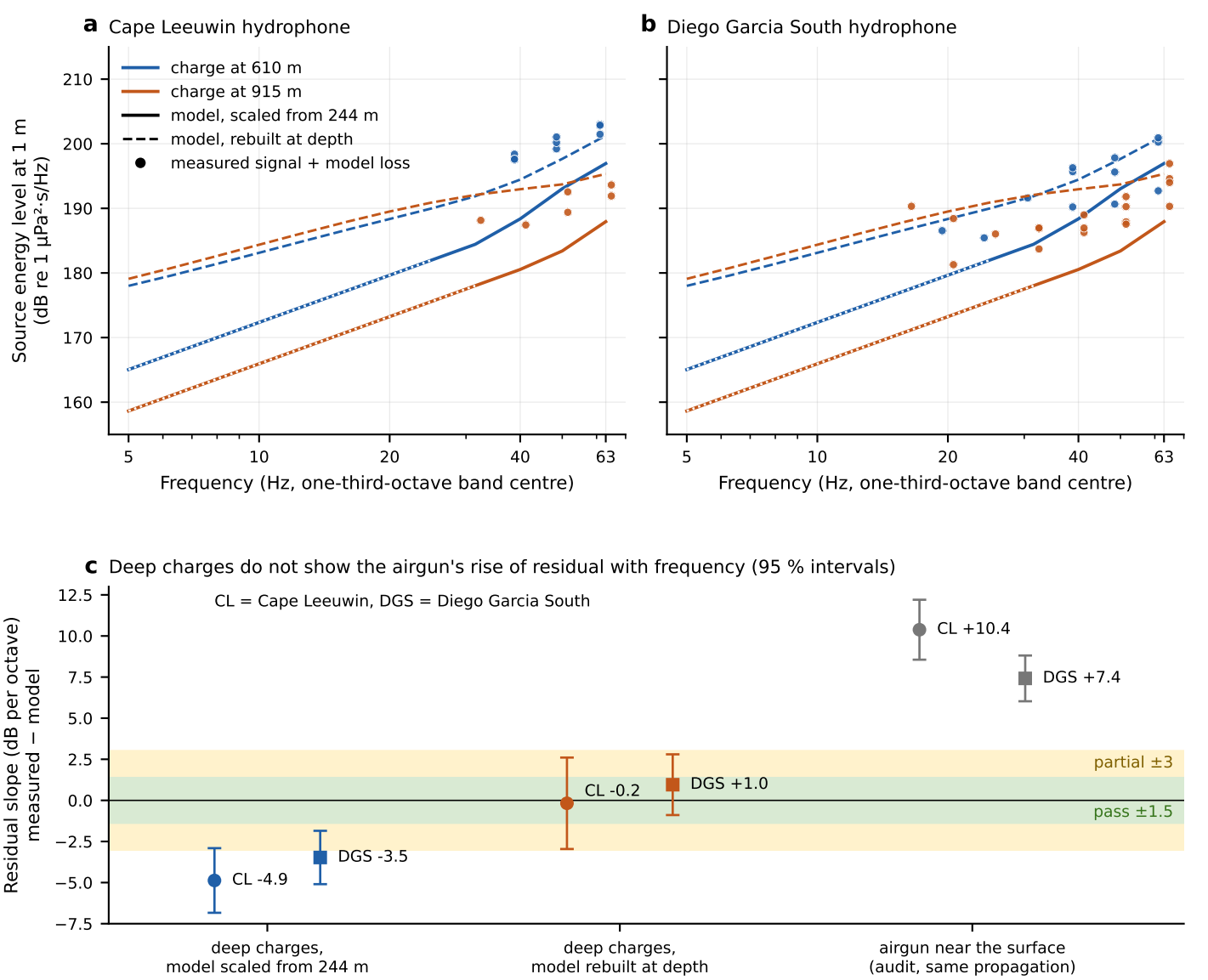

STE100: UNCONVERGED (the core filter is not converged; this test does not use it). Panels a and b: the dots are the level that the charge had to make at 1 m, from the measured signal and the model loss. The lines are two estimates of that level from Gaspin and Shuler (1971). The solid line scales their 244 m result to the true depth. The broken line builds the pulse again at the true depth. The white dots on a line show where the scaled estimate goes below their lowest band. The two estimates are different by 4 to 17 dB in the bands that we use, so the absolute level is not known well. Panel c: the frequency slope of measured minus model. The deep charges have a slope near zero or negative. The airgun near the surface had a large positive slope.

Technical: core (b) split-half NOT CONVERGED (not used). Source: G&S NOLTR 71-160 Table 3 (p. 12), 1.8 lb TNT, free field, ideal 1/3-oct bands, dB re 1 erg/cm²/Hz at 100 yd, converted with ρc = 1.539e5 cgs and 20 log 91.44 m. Solid: time similitude from 800 ft, s = (833/(z+33))^(5/6) ft (G&S §7.5); dotted = extrapolated at the 12.5–16 Hz slope. Dashed: Table 2 construction re-implemented at depth (κ = 4, sin² negative phases; reproduces Table 3 at 800 ft to RMS 1.3 dB). Both outside G&S's stated validity (≤ ~1000 ft, §8.1). Points: audit implied SL = S_b + TL_b + 10 log 6 s, bands with (S+N)/N ≥ 3 dB only; TL = module KRAKEN wrapper on audit paths (WOA23, GEBCO), band centre. 8 shots of 1.8 lb (A6–A10), 12 shot-station pairs, 45 bands, 16–63 Hz; A11 excluded (charge not 1.8 lb). Panel c: within-pair OLS slope of r_b on log2 f, pooled per station, ±1.96 SE; air9 from audit C1 (12.5–63 / 6.3–63 Hz). Shading = C4 slope thresholds (criteria.md, 940f1a3).